

Divyansh Gupta

DATA SCIENTIST

Tempe, AZ, USA | (928) 408-6668 | divguptaa81@gmail.com | <http://www.linkedin.com/in/divyanshgupttaa>

SUMMARY

- Data Scientist with 3+ years of experience developing AI-driven healthcare and financial analytics solutions, specializing in imaging intelligence, predictive modeling, and enterprise-scale data systems using **Python**, **TensorFlow**, and **PyTorch**.
- Skilled in building production-grade machine-learning pipelines, deploying cloud-hosted inference services using **AWS SageMaker**, **EC2**, and **S3**, and enhancing model performance with advanced evaluation workflows and multimodal data engineering.
- Proven ability to deliver measurable business impact—accelerating diagnostic workflows, improving risk-assessment accuracy, and optimizing data operations—by designing scalable models, automation frameworks, and high-availability analytical systems.

SKILLS

Languages:	Python, SQL, R
ML & AI:	Scikit-learn, TensorFlow/Keras, Deep Learning, RNN/LSTM, NLP, Predictive Modeling, LangGraph, Crew AI
GenAI & LLM:	Large Language Models, Retrieval-Augmented Generation, Model Fine-Tuning, Agentic AI Frameworks
Statistical Methods:	Regression Analysis, Time Series Analysis, Bayesian Methods, A/B Testing, Hierarchical Modeling
Model Evaluation:	DeepEval, RAGAS, LLM Performance Assessment, Model Validation Frameworks
Cloud & Big Data:	AWS concepts, Azure AI Services, Large-Scale Data Processing, Scalable Data Pipelines, Cloud ML Tech
Visualization & BI:	Tableau, Power BI, Advanced Excel, Dashboard Development, Data Storytelling

WORK EXPERIENCE

Data Scientist | McKesson Corporation, USA

Aug 2024 – Present

- Led a multi-site clinical imaging intelligence program for 7 radiology groups, architecting systematized diagnostic workflows using **TensorFlow** and enterprise-grade imaging pipelines, increasing radiologist throughput across high-volume environments.
- Designed CNN-driven anomaly-detection workflows trained on 120K radiology images using **Keras**, generating PACS-compatible heatmaps that reduced interpretation time and improved clinical review consistency across distributed imaging centers.
- Expanded training datasets by synthesizing 18K rare-disease examples with **Stable Diffusion** and **PyTorch**, strengthening class balance across 12 imaging modalities and improving detection reliability for low-prevalence pathologies.
- Accelerated ingestion and transformation of 50K+ DICOM studies by implementing distributed processing across **Apache Spark** and **Hadoop**, reducing imaging-pipeline latency and enabling near-real-time QA review for 5 hospital regions.
- Computerized radiology reporting workflows using **Gemini API** and **Whisper API**, reducing manual dictation burden for 400 clinicians and improving documentation turnaround time by 34% across enterprise imaging operations.
- Deployed scalable inference endpoints using **AWS SageMaker**, **EC2**, **S3**, and **ECR**, enabling secure, HIPAA-aligned model hosting with low-latency access and improving overall diagnostic-decision support availability by 27%.

Data Scientist | Deloitte, India

Jan 2021 - Jun 2023

- Modernized credit-risk modeling workflows for 11 lending units by developing predictive scoring pipelines using **Python**, **scikit-learn**, and **RNN/LSTM**, improving early detection of high-risk borrowers across sub-prime segments.
- Processed and analyzed 1.8M+ borrower repayment events using optimized **SQL** and **Oracle**, integrating **FAISS** for similarity-based cohort segmentation to enhance risk grouping accuracy across diversified portfolios.
- Built portfolio volatility and delinquency prediction models leveraging **Bayesian Inference** and **Time-Series Analysis**, delivering proactive alerts used by 20+ senior portfolio managers through dashboards in **Power BI**.
- Deployed secured risk-scoring microservices with **FastAPI**, containerized using **Docker**, and hosted on **AWS**, ensuring consistent high-availability predictions across multi-region financial infrastructure.
- Conducted hierarchical A/B experiments using advanced **statistical testing** techniques to calibrate risk thresholds, reducing false-positive alerts by 21% and improving screening precision for corporate and SME segments.
- Automated feature engineering pipelines using **Python** and **Pandas**, reducing manual data preparation time by 36% and standardizing variable definitions across underwriting, fraud, and collections teams.
- Implemented end-to-end model monitoring using **scikit-learn** evaluation metrics and drift-detection scripts, ensuring regulatory-compliant model stability across quarterly audit cycles.
- Collaborated with cross-functional credit, compliance, and actuarial teams to map lending policies to model outputs, integrating insights into decisioning tools powered by **SQL**, **Power BI**, and secure **AWS** endpoints.

EDUCATION

Master of Science in Information Technology | Arizona State University, Tempe, AZ

Aug 2023 - May 2025

Bachelor of Engineering in Information Technology | Nagpur University, Nagpur, India

Jul 2017 - Jul 2021